

An Alternative Method for Edge Detection Using Schwarzian Derivatives

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Abstract

The performance of edge detection techniques plays a crucial role in various image processing applications. In this study, a new perspective on traditional filters is introduced by utilizing Schwarzian derivatives, representing the first application of an edge detection filter based on Schwarzian derivatives [1]. Experimental results show that the quantitative performance of the proposed algorithm does not create a significant difference compared to traditional methods such as Sobel [2], Prewitt [3] and Canny [4]; however, it has been observed that it produces visual results with higher contrast and more distinct edges. The evaluation uses PSNR, SSIM, and MSE, confirming the effectiveness of the proposed approach.

Keywords: Edge detection, Schwarzian derivatives, image processing, Sobel operator, Prewitt operator, Canny operator, Otsu thresholding.

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